

Sliding Window

originally $\rightarrow$ networking $\rightarrow$ protocols.

Problem solving $\Rightarrow$ optimise solⁿ.

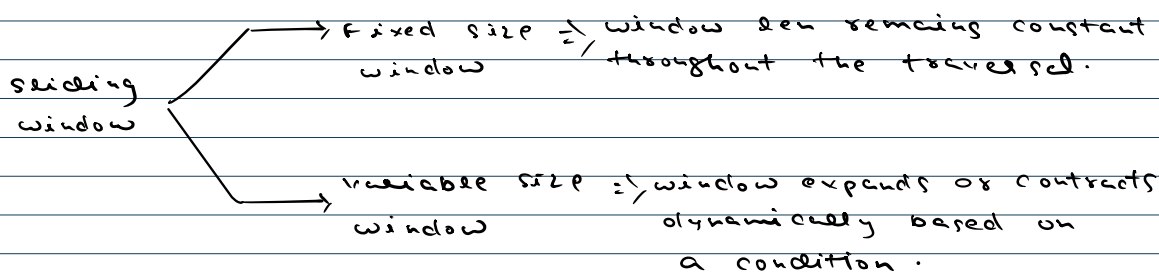
When to apply?

Input $\Rightarrow$ Array / String

k (int) {sliding window size}

Output $\Rightarrow$ Subarray / Substring

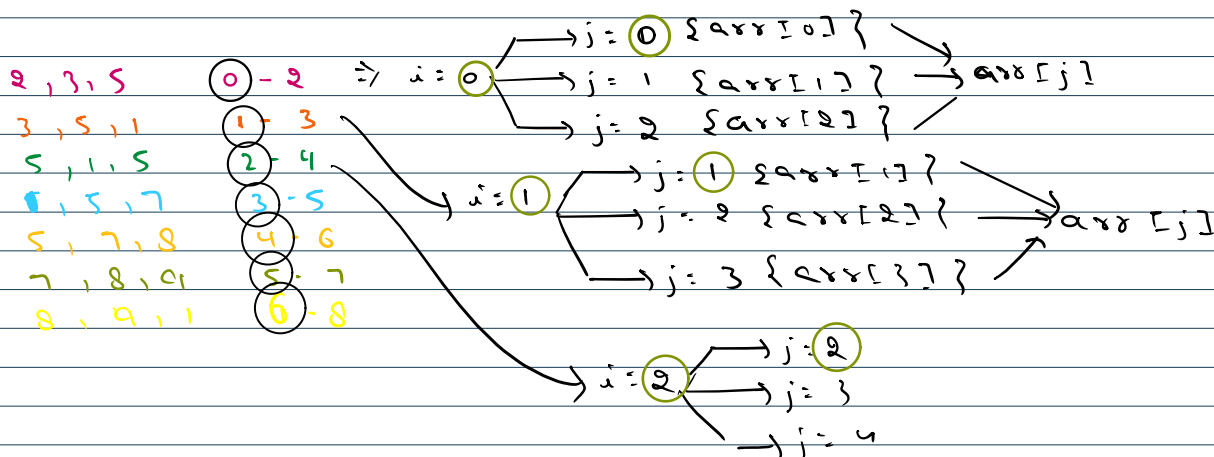
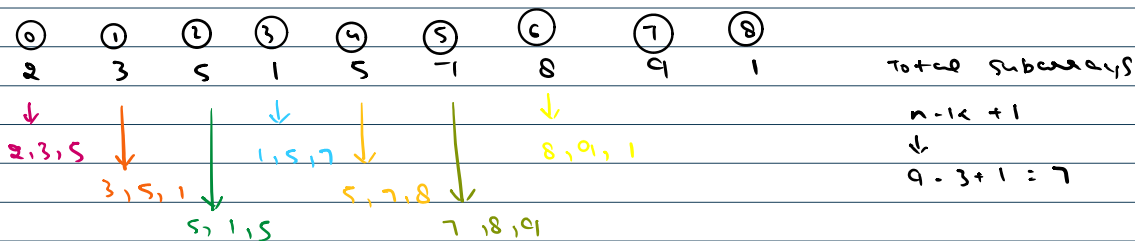
cmdth $\Rightarrow$ max, min, sum, count, frequency



Maximum sum of k consecutive elements

arr $\rightarrow$ { 2, 3, 5, 1, 5, 7, 8, 9, 1 } Input
 $k \rightarrow 3$
 OP

Output $\rightarrow$ max sum of any contiguous subarray size of k .



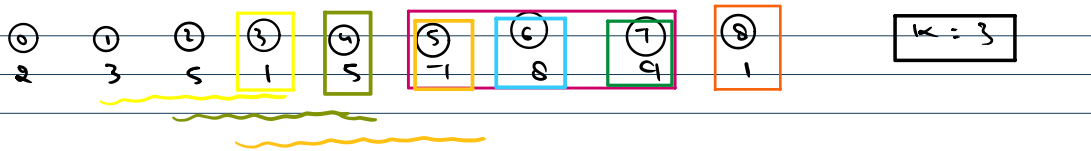
$i = 0$ to $n - k$
 $j = i$ to $i + k$

Sliding Window technique

Fixed Size

1st window calculate

{
 // Grow
 // Shrink
 // Update the answer

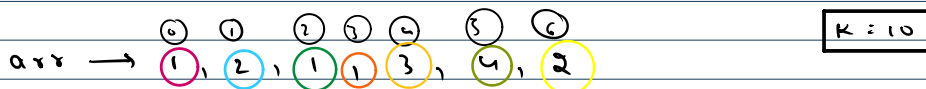


1st window $\Rightarrow 2, 3, 5 \Rightarrow 2 + 3 + 5 = 10$

	add	sub	
sum = 10 + 1 = 11 - 2 = 9	arr[3]	arr[0]	$\Rightarrow 2, 3, 5, 1$
9 + 5 = 14 - 3 = 11	arr[4]	arr[1]	$\Rightarrow 3, 5, 1, 5$
11 + 7 = 18 - 5 = 13	arr[5]	arr[2]	$\Rightarrow 5, 1, 5, 7$
13 + 8 = 21 - 1 = 20	arr[6]	arr[3]	$\Rightarrow 1, 5, 7, 8$
20 + 9 = 29 - 5 = 24	arr[7]	arr[4]	$\Rightarrow 5, 7, 8, 9$
24 + 1 = 25 - 7 = 18	arr[8]	arr[5]	$\Rightarrow 7, 8, 9, 1$
max ↓ Answer	arr[i] i: k to n-1	arr[i-k] i: k to n-1	

Subarray Product Less Than K

```
loop {
  // Grow
  // Shrink
  // update answer
}
```



1 2 1 1 3 4 2
 1 2 2 1 1 3 4 2
 1 2 1 2 1 1 3 4 2
 1 2 1 1 2 1 1 3 4 2
 1 2 1 1 3

Grow → Product
 Shrink → Divide

arr → 0 1 2 3 4 5
 1, 2, 1, 3, 4, 2

$k = 10$

1 2 1 3 4 2
 1 2 2 1 1 3 4 2
 1 2 1 2 1 1 3 4 2
 1 2 1 1 2 1 1 3 4 2
 1 2 1 1 3

$$1 + 2 + 3 + 4 + 1 + 2 = 13$$

$k = 10$

Shrink → Divide
 Grow → Product

$s = 0$
 $e = 0$

arr → 0 1 2 3 4 5
 1, 2, 1, 3, 4, 2

$e = 2, s = 0$
 $size = e - s + 1$
 $e = 3, s = 0$
 $size = e - s + 1$

$prod = 1 \times 2 \times 1 \times 3 \times 4 \times 2$
 $e = 0 \Rightarrow prod * arr[0]$
 $e = 1 \Rightarrow prod * arr[1]$
 $e = 2 \Rightarrow prod * arr[2]$
 $e = 3 \Rightarrow prod * arr[3]$
 $e = 4 \Rightarrow prod * arr[4]$
 $e = 5 \Rightarrow prod * arr[5]$

$$\begin{aligned}
 prod &= 24 : 24 \\
 &= \frac{24}{1} : 12 \\
 &= \frac{12}{2} : 6 \\
 &= \frac{6}{3} : 2 \\
 &= \frac{2}{4} : 0.5
 \end{aligned}$$

$0 \text{ to } n - 1 \Rightarrow \text{loop}$
 $Window = end - start + 1$
 $size$

Arrays - Sum of Two Arrays

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